Purdue University **2026-2027 University Catalog**

Computer Engineering, BSCMPE

Location(s): West Lafayette, Indianapolis

About the Program

The BS program in Computer Engineering is accredited by the Engineering Accreditation Commission of ABET, under the commission's General Criteria and Program Criteria for Electrical, Computer, Communications, Telecommunication(s) and Similarly Named Engineering Programs.

Electrical and Computer engineering encompasses all areas of research, development, design, and operation of electrical and electronic systems and their components, including software. Emphasis in such varied areas as artificial intelligence and machine learning, bioengineering, circuit theory, very large-scale integrated circuits, communication sciences, computer engineering and embedded systems, software development, control and robotic systems, electromagnetic fields, energy sources and systems, quantum technology, and semiconductors are available. Two degree programs are offered by the Elmore Family School of Electrical and Computer Engineering at both West Lafayette and Indianapolis locations: Bachelor of Science in Electrical Engineering (BSEE) and Bachelor of Science in Computer Engineering (BSCmpE).

Graduates from the Elmore Family School of Electrical and Computer Engineering are sought after in every industry, in business/finance, and in government. They launch startups and lead corporations at the highest levels. Electrical and Computer engineers hold many challenging positions in the aerospace, automotive, chemical, nuclear, medical, metallurgical, textile, railway, petroleum, and other basically non-electrical industries, as well as in computers, electronics, communications, power, and other traditional electrical industries. Their professional roles span management, industrial activity, research, development, design, production, marketing, operation, field testing, and maintenance.

Electrical and Computer Engineering Major Change (CODO) Requirements

Elmore Family School of Electrical and Computer Engineering

Degree Requirements

125 Credits Required

Computer Engineering Required Major Courses (53 credits)

ECE requires a minimum of a 2.0 for major GPA. All Computer Engineering Core Courses, Seminar Courses, Senior Design, and Computer Engineering Selective courses count towards the Computer Engineering major GPA calculation.

Computer Engineering Core Courses (30 credits)

- ECE 20001 - Electrical Engineering Fundamentals I **Credit Hours: 3.00** (minimum grade of C)
- ECE 20007 - Electrical Engineering Fundamentals I Lab **Credit Hours: 1.00**
- ECE 20002 - Electrical Engineering Fundamentals II **Credit Hours: 3.00** (minimum grade of C)
- ECE 26400 - Advanced C Programming **Credit Hours: 3.00** (minimum grade of C)
- ECE 20875 - Data Science Using Python **Credit Hours: 3.00**
- ECE 27000 - Introduction To Digital System Design **Credit Hours: 4.00** (minimum grade of C)
- ECE 30100 - Signals And Systems **Credit Hours: 3.00**
- ECE 30200 - Probabilistic Methods In Electrical And Computer Engineering **Credit Hours: 3.00**
- ECE 36200 - Microprocessor Systems And Interfacing **Credit Hours: 4.00**
- ECE 36800 - Data Structures **Credit Hours: 3.00**

Electrical and Computer Engineering Seminar Courses (3 credits)

- ECE 29401 - Electrical And Computer Engineering Sophomore Seminar **Credit Hours: 1.00**
- ECE 39401 - Communication & Collaboration **Credit Hours: 1.00**
- ECE 49401 - Professional Communication Capstone **Credit Hours: 1.00**

Senior Design Requirement - Choose One Option (4 credits)

Students must complete all Computer Engineering Core Courses before taking senior design.

- ECE 47700 - Digital Systems Senior Project **Credit Hours: 4.00**

  OR
- ECE 49022 - Electrical Engineering Senior Design Projects **Credit Hours: 4.00**

  OR
- EPCS 41201 - Senior Design Participation In EPICS I **Credit Hours: 2.00** and
- EPCS 41202 - Senior Design Participation In EPICS II **Credit Hours: 2.00**

  OR
- VIP 47921 - Senior Design Participation In Vertically Integrated Projects (VIP) I **Credit Hours: 2.00** and
- VIP 47922 - Senior Design Participation In Vertically Integrated Projects (VIP) II **Credit Hours: 2.00**

Computer Engineering Selectives (16 credits)

No more than 6 credit hours of Computer Engineering Special Content courses can count towards Computer Engineering Selectives.

See Computer Engineering Selectives

Other Departmental/Program Course Requirements (66-82 credits)

Other Course Requirements (42-55 credits)

Intro to Engineering I & II - Credit Hours: 4.00-8.00

- ENGR 13100 - Transforming Ideas To Innovation I (UCC: AI/IL) **Credit Hours: 2.00** and
- ENGR 13200 - Transforming Ideas To Innovation II (UCC: AI) **Credit Hours: 2.00**

OR

- EPCS 11100 - First Year Participation In EPICS I (UCC: STS) **Credit Hours: 1.00** and
- EPCS 12100 - First Year Participation In EPICS II (UCC: STS) **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) **Credit Hours: 2.00**

OR

- VIP 17911 - First Year Participation In Vertically Integrated Projects (VIP) I (UCC: STS) **Credit Hours: 1.00** and
- VIP 17912 - First Year Participation In Vertically Integrated Projects (VIP) II **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) - **2.00** **Credits Hours**

OR

- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) **Credit Hours: 4.00** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) **Credit Hours: 4.00**

OR

- ENGR 13000 - Transforming Ideas Into Innovations (UCC: AI/IL) **Credit Hours: 4.00**

- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

AND

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- CS 15900 - C Programming **Credit Hours: 3.00** (minimum grade of C-)

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

Mathematics Requirement - Credit Hours: 13.00-14.00

- ECE 20869 - Discrete Math For Computer Engineering **Credit Hours: 3.00**
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**

- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00** and
- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00**
  OR
- MA 26200 - Linear Algebra And Differential Equations (UCC: QR) **Credit Hours: 4.00** and
- Advanced Math Selective - Credit Hours: 3.00 (see supplemental information)

Physics I - Credit Hours: 0.00-4.00

- PHYS 17200 - Modern Mechanics (UCC: SCI) **Credit Hours: 4.00**
  OR
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) - **4.00 Credit Hours** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) - **4.00 Credit Hours**

- Engineering Breadth Selective - Credit Hours: 3.00
- Science Requirement - Credit Hours: 7.00-8.00

Engineering General Education Requirements (24-25 credits)

For a list of courses that satisfy this requirement, see Supplemental Information

- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00
- General Education II (UCC: BSS) - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00
- Written Communication (UCC: WC) - Credit Hours: 3.00-4.00
- Oral Communication (UCC: OC) - Credit Hours: 3.00

Additional General Education Requirements:

- *At least 6 credits must be in non-introductory courses (30000-level or above OR from courses with a required pre-requisite in the same department)*
- *At least 12 credits must be taken from the College of Liberal Arts, the Daniels School of Business, Psychological Sciences and/or the Honors College provided such courses are not focused primarily on engineering, technology, the natural sciences, or mathematics.*
- *A course is not eligible for use towards the 24 credits of Engineering General Education requirement if that course is used to satisfy a different engineering degree requirement.*

Electives (0-2 credits)

- Electives - Credit Hours: 0.00-2.00

Supplemental Information

- Computer Engineering Selectives
- Electrical & Computer Engineering Supplemental Information

Optional Concentrations

- Automatic Control Concentration for Computer Engineering & Electrical Engineering (ACON)
- Artificial Intelligence and Machine Learning Concentration for Computer Engineering (AICE)
- Computer Systems Concentration in Computer Engineering & Electrical Engineering (CSYS)
- Electric Power and Energy Systems Concentration for Computer Engineering & Electrical Engineering (EPES)
- Microelectronics and Semiconductor Concentration for Computer Engineering (MSCE)
- Quantum Technology Concentration for Computer Engineering & Electrical Engineering (QNTM)
- RF & Optical Engineering Concentration for Computer Engineering & Electrical Engineering (RFOE)
- Software Engineering Concentration for Computer Engineering & Electrical Engineering (SFTE)

GPA Requirements

- A cumulative GPA of 2.0 is required for all Baccalaureate degrees.
- ECE requires a minimum of a 2.0 for major GPA. All Computer Engineering Core Courses, Seminar Courses, Senior Design, and Selective courses count towards the Computer Engineering major GPA calculation.

Pass/No Pass Policy

- The pass/no pass (P/NP) grade option, if available, may be used for courses taken to satisfy the Engineering General Education Requirement and Elective credits. The P/NP grade option cannot be used for courses applied towards the Required Major Courses or Other Course Requirements sections (unless P/NP is the only allowed grade option for that course).

Notes

- Students in Computer Engineering, BS cannot also enroll in the Computer Science, BS, Computer Science minor, or the Electrical and Computer Engineering Minor.

Transfer Credit Policy

If you are interested in registering for a course offered by a different institution, please review the information on the Purdue Office of the Registrar Transfer Credit website.

* If you are an admitted transfer student, please review the above Transfer Credit website and then work with your academic advisor to determine exactly how your previous courses might apply.

University Requirements

University Core Requirements

FOR A COMPLETE LISTING OF UNIVERSITY CORE COURSE SELECTIVES, VISIT THE UNIVERSITY SENATE WEBSITE.

- AI Working Competency (UCC: AI)
- Human Cultures: Behavioral/Social Science (UCC: BSS)
- Human Cultures: Humanities (UCC: HUM)
- Information Literacy (UCC: IL)
- Oral Communication (UCC: OC)
- Quantitative Reasoning (UCC: QR)
- Science #1 (UCC: SCI)
- Science #2 (UCC: SCI)
- Science, Technology, and Society (UCC: STS)
- Written Communication (UCC: WC)

Civics Literacy Proficiency Requirement

The Civics Literacy initiative is designed to develop civic knowledge of Purdue students to graduate a more informed citizenry. For more information visit the Civics Literacy - Office of the Vice Provost for Teaching and Learning - Purdue University.

To obtain the Civics Literacy credential, all Baccalaureate degree seeking students will complete the Civics Literacy Podcast pathway - Complete 12 podcasts created by the Purdue Center for C-SPAN Scholarship & Engagement that use C-SPAN material and pass the required quizzes per episode. After the pathway has been successfully completed, students will take the Civics Literacy knowledge test. Civics Literacy does not impact a student's GPA or bear any financial costs.

More details about each pathway and how to complete the requirement can be found on the Civics Literacy Student Dashboard in myPurdue.

Upper Level Requirement

- Resident study at Purdue University for at least two semesters and the enrollment in and completion of at least 32 semester hours of coursework required and approved for the completion of the degree. These courses are expected to be at least junior-level (30000+) courses.
- Students should be able to fulfill *most*, *if not all*, of these credits within their major requirements; there should be a clear pathway for students to complete any credits not completed within their major.

Sample Plan of Study

The following sample plan of study is a suggested arrangement of courses. Consultation with an advisor may result in an altered plan customized for an individual student.

Fall 1st Year

- Intro to Engineering I – Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

13-17 Credits

Spring 1st Year

- Intro to Engineering II - Credit Hours: 2.00-4.00

- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- Physics I - Credit Hours: 4.00
- CS 15900 - C Programming **Credit Hours: 3.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

16-20 Credits

Fall 2nd Year

- ECE 29401 - Electrical And Computer Engineering Sophomore Seminar **Credit Hours: 1.00**
- ECE 20001 - Electrical Engineering Fundamentals I **Credit Hours: 3.00**
- ECE 20007 - Electrical Engineering Fundamentals I Lab **Credit Hours: 1.00**
- ECE 20869 - Discrete Math For Computer Engineering **Credit Hours: 3.00**
- ECE 26400 - Advanced C Programming **Credit Hours: 3.00**
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**

15 Credits

Spring 2nd Year

- ECE 20002 - Electrical Engineering Fundamentals II **Credit Hours: 3.00**
- ECE 20875 - Data Science Using Python **Credit Hours: 3.00**
- ECE 27000 - Introduction To Digital System Design **Credit Hours: 4.00**

- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00** or
- MA 26200 - Linear Algebra And Differential Equations (UCC: QR) **Credit Hours: 4.00**

- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00

16-17 Credits

Fall 3rd Year

- ECE 30100 - Signals And Systems **Credit Hours: 3.00**
- ECE 36200 - Microprocessor Systems And Interfacing **Credit Hours: 4.00**
- ECE 36800 - Data Structures **Credit Hours: 3.00**
- ECE 39401 - Communication & Collaboration **Credit Hours: 1.00**
- Science Selective - Credit Hours: 3.00-4.00
- General Education II (UCC: BSS) - Credit Hours: 3.00

17-18 Credits

Spring 3rd Year

- ECE 30200 - Probabilistic Methods In Electrical And Computer Engineering **Credit Hours: 3.00**
- PHYS 27200 - Electric And Magnetic Interactions (UCC: SCI) **Credit Hours: 4.00**
- Computer Engineering Selectives - Credit Hours: 7.00
- General Education III - Credit Hours: 3.00

17 Credits

Fall 4th Year

- Senior Design Requirement - Credit Hours: 2.00-4.00
- Computer Engineering Selectives - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00
- Elective - Credit Hour: 0.00-2.00

- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00** or
- Advanced Math Selective - Credit Hours: 3.00

13-14 Credits

Spring 4th Year

- ECE 49401 - Professional Communication Capstone **Credit Hours: 1.00**
- Senior Design Requirement - Credit Hours: 0.00-2.00
- Computer Engineering Selectives - Credit Hours: 6.00

- Engineering Breadth Selective - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00

16-18 Credits

Pre-Requisite Information

For pre-requisite information, log in to mypurdue.purdue.edu and click here.

World Language Courses

World Language proficiency requirements vary by program. The following list is inclusive of all world languages PWL offers for credit; for acceptable languages and proficiency levels, see your advisor. (ASL-American Sign Language; ARAB-Arabic; CHNS-Chinese; FR-French; GER-German; GREK-Greek(Ancient); HEBR-Hebrew(Biblical); HEBR-Hebrew(Modern); ITAL-Italian; JPNS-Japanese; KOR-Korean; LATN-Latin; PTGS-Portuguese; RUSS-Russian; SPAN-Spanish)

Critical Course

The ♦ course is considered critical.

In alignment with the Degree Map Guidance for Indiana's Public Colleges and Universities, published by the Commission for Higher Education (pursuant to HEA 1348-2013), a Critical Course is identified as "one that a student must be able to pass to persist and succeed in a particular major. Students who want to be nurses, for example, should know that they are expected to be proficient in courses like biology in order to be successful. These would be identified by the institutions for each degree program."

Disclaimer

The student is ultimately responsible for knowing and completing all degree requirements. Consultation with an advisor may result in an altered plan customized for an individual student. The myPurduePlan powered by DegreeWorks is the knowledge source for specific requirements and completion.

Comparative information about Purdue University and other U.S. educational institutions is also available through the College Navigator tool, provided by the National Center for Education Statistics, and through the U.S. Department of Education College Scorecard.